Multimodal biometric authentication: A Review

Noah Wiley, Nick Martinez,
Xavier Maravilla

Introduction

- Unimodal systems widely used
- Weak to noise, spoofing, environmental issues
- No fallback if template compromised
- Multimodal improves reliability

Key Motivation

- Need Balanced FAR/FRR
- Unimodal complexity grows with feature count
- Multimodal reduces ambiguity
- Increased robustness against spoofing

Related Work: Part 1

- Issues with prior surveys:
  - Too narrow in scope
  - Focused on one modality
  - Did not address security
  - Ignored dataset diversity

Related Work: Part 2

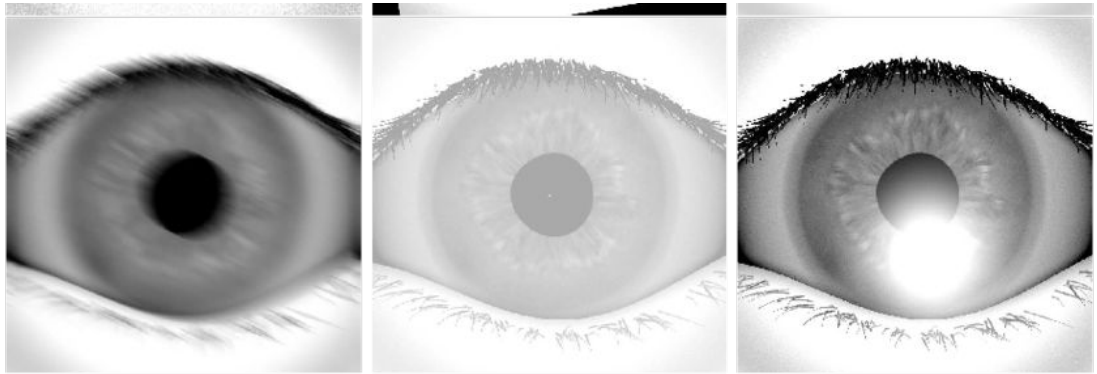
- More gaps in earlier studies:
  - No template protection
  - Lack of multimodal focus
  - No discussion of privacy

TABLE I
COMPARISON OF RECENT BIOMETRIC AUTHENTICATION SURVEYS

Authors	Main Focus	Limitations / Gaps Identified
Shaheed et al. (2021)	Systematic review of physiological-based biometric traits (hand, face, ocular).	Lacked analysis of authentication mechanisms and privacy protection.
Shaveta et al. (2020)	Traits-based discussion of behavioral and physical identifiers.	Did not include datasets or security considerations.
Minaee et al. (2023)	Application of deep learning techniques in biometric recognition.	No discussion of authentication protocols or dataset diversity.
Abdulrahman et al. (2023)	Comparison of five physiological and behavioral traits.	Did not examine multimodal integration or template protection.
Bala et al. (2022)	Review of multimodal systems using fusion techniques.	No mention of privacy, data integration, or template revocability.
Uwaechia et al. (2021)	ECG-based biometric authentication.	Focused on a single modality; no cross-modality evaluation.
Pahuja & Goel (2024)	Comprehensive review covering authentication methods, datasets, and multimodal systems.	Addresses prior gaps in template protection, data privacy, and system integration.

Problem Statement

- **Biometric weaknesses:**
  - **Fingerprints → smudges, spoofing**
  - **Face → lighting, expression**
  - **Iris → motion discomfort**
  - **Single trait = Single point of failure**

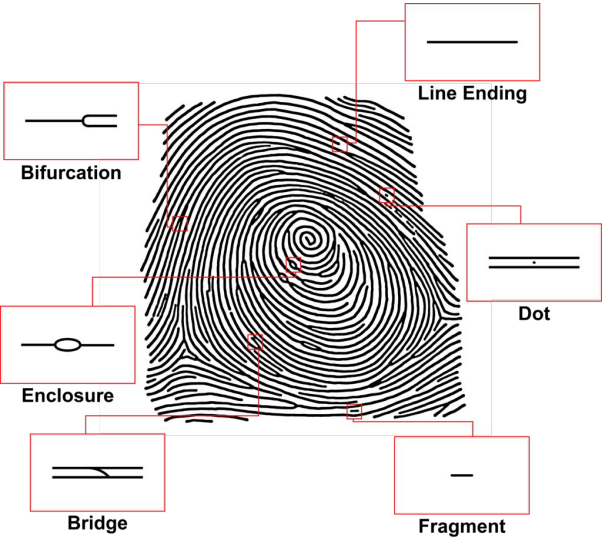


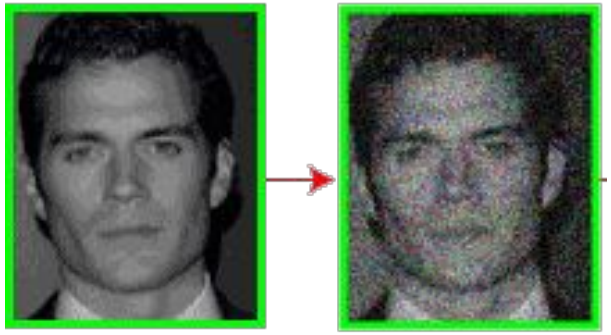
Methodology Overview

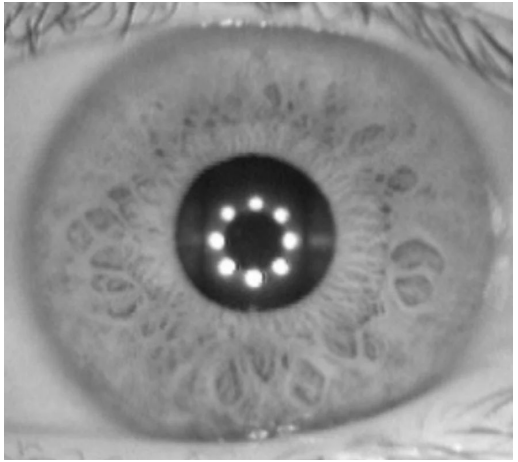
- Four major components:
  - Data Acquisition
  - Preprocessing
  - Feature Extraction
  - Fusion + Classification

Feature Extraction

- Fingerprint → minutiae, ridge orientation, texture
- Face → LBP, CNN embeddings
- Iris → Gabor wavelets, phase encoding







Limitations of Current Authentication

- Traditional Methods are Obsolete
- Unimodal (Single Trait) Systems are Unreliable
- High Error Rates (FAR/FRR)
- Vulnerable to Spoofing Attacks
- High Failure To Enroll (FTE) Rate

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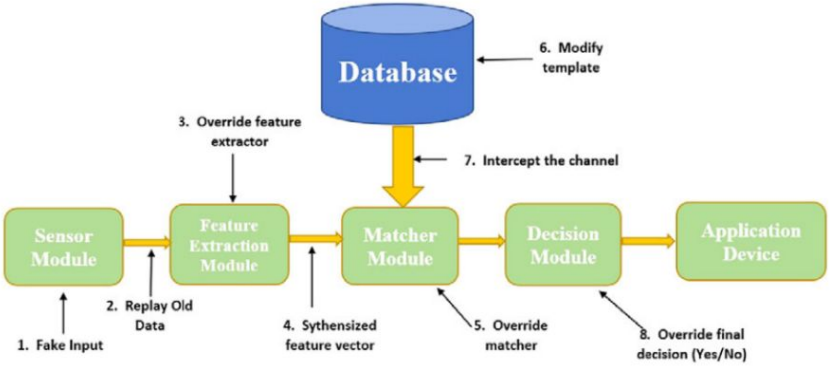
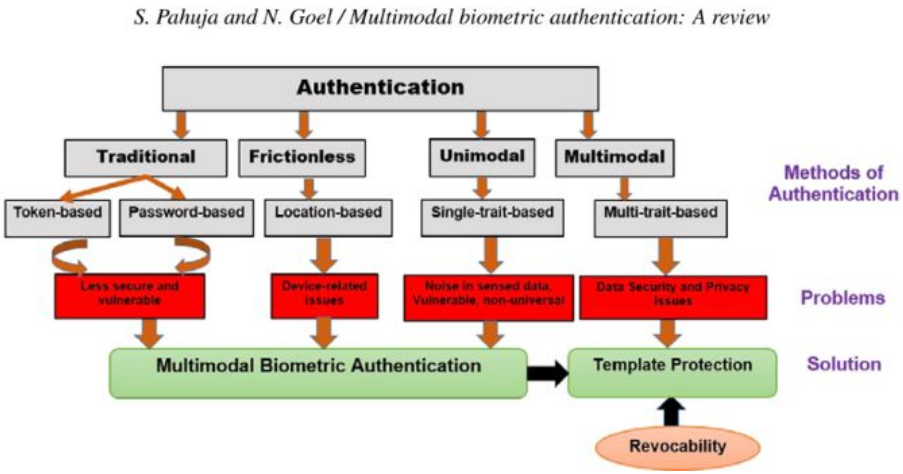


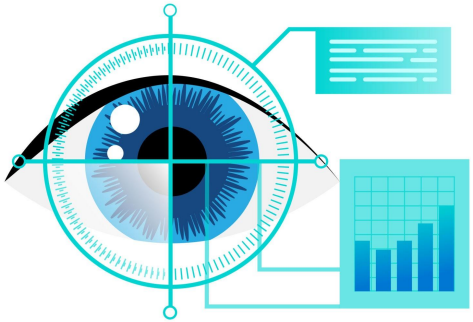
The Core Problem: The Non-Revocable Template



- The "Accuracy Fix": Multimodal Systems
- The Unsolved, Critical Flaw: Template Security
- The Non-Revocable Crisis
- The Failure of Current Solutions
- The "Security vs. Accuracy" Trade-Off

Proposed Solution: A Revocable & Robust Framework

- **Component 1: Multimodal Fusion (Fingerprint + Iris)**
- **Complementary Trait Rationale**
- **Component 2: Integrated Revocable Templates**



Thank You
Questions?